

Alternative proof

$$\# \text{ multisets} = \binom{n+r-1}{n-1}$$

$n+r-1$ cells. Colour $n-1$ cells blue to show where we switch to next element

From multiset to colouring

Start with $s_i = s_1$

Repeat If c_i copies of s_i : ship c_i steps right
 then colour next cell blue
 update s_i to s_{i+1} could be 28720

We colour $n-1$ cells } $n+r-1$
 We ship total of r cells }

Different multisets $\Rightarrow$ different colourings
 (look at lowest s_i for which there is difference)

Shows that all multisets give rise to colourings/
 subset choices. Now show that all colourings/
 subset choices give multisets

From colouring to multiset

Set $s_i := s_1$

for position = leftmost cell up to rightmost

~~for~~ if position blue

$s_i := s_{i+1}$

else

add copy of s_i to set
 [move position right]

Ex What is the number of possible different outcomes when rolling 3 (indistinguishable) dice? CVII

$$n = 6$$

$$r = 3$$

outcomes = # 3-element multisets

$$= \binom{6+3-1}{3} = \frac{8!}{3!5!} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56$$

SUMMING UP what we did so far
n elements; choose r times

	WITH REPETITIONS	WITHOUT REPETITIONS
ORDERED	sequence n^r	permutation $nPr = \frac{n!}{(n-r)!}$
UNORDERED	multiset $n+r-1Cr = \binom{n+r-1}{r}$	set $nCr = \binom{n}{r}$

BINOMIAL COEFFICIENT

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

"n choose r"

EXAMPLE COMBINATORIAL PROOF

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

How to prove?

Proof 1

Just calculate. Work on right-hand side

$$\begin{aligned}\binom{n-1}{k-1} + \binom{n-1}{k} &= \frac{(n-1)!}{(k-1)!(n-k)!} + \frac{(n-1)!}{k!(n-k-1)!} \\&= \frac{k \cdot (n-1)!}{k \cdot (k-1)!(n-k)!} + \frac{(n-k)(n-1)!}{(n-k)k!(n-k-1)!} \\&= \frac{(k+n-k)(n-1)!}{k!(n-k)!} = \frac{n!}{k!(n-k)!} \\&= \binom{n}{k}\end{aligned}$$

Proof 2 (combinatorial)

Right-hand side is talking about different ways of choosing subsets from set of $n-1$ elements...

Left-hand side talks about n elements.

Make case analysis for choosing k elements out of n

(a) n th element is in chosen set

(b) n th element is not in chosen set

In case (a), have to choose $k-1$ elements from remaining $n-1$ elements in $\binom{n-1}{k-1}$ ways

In case (b), have to choose all k elements from remaining $n-1$ elements in $\binom{n-1}{k}$ ways

These are all ways of choosing k elements out of n , so

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

SOME PROBABILITY THEORY

| Pr |

(DISCRETE) SAMPLE SPACE: set of outcomes

Rolling a dice: $A = \{1, 2, 3, 4, 5, 6\}$

Probability function $p: A \rightarrow \mathbb{R}$

$$p(a) \geq 0 \quad \forall a \in A$$

$$\sum_{a \in A} p(a) = 1$$

also denoted
 $\Pr[a]$

(p, A) is a PROBABILITY SPACE

EVENT $E \subseteq A$ subset of sample space

Rolling an even number $E = \{2, 4, 6\}$

$$p(E) = \sum_{a \in E} p(a)$$

When all events equally likely, probabilities reduce to counting

$$p(E) = \sum_{a \in E} p(a) = \sum_{a \in E} \frac{1}{|A|} = \frac{|E|}{|A|}$$

number of outcomes in E
total # outcomes

Ex Probability of drawing 3 kings from deck of cards when drawing 3 cards

All outcomes equally likely

A = all possible 3-card hands $|A| = {}_{52}C_3 = \binom{52}{3}$

E = only kings $|E| = 4$ (why?)

$$p(E) = \frac{4}{\binom{52}{3}} = \frac{4}{22100} = \frac{1}{5525}$$

Set operations for events

Pr II

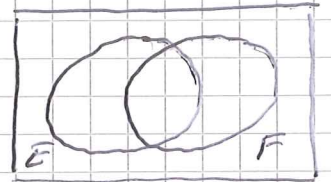
UNION $E \cup F = \{a \in A \mid a \in E \text{ or } a \in F\}$

INTERSECTION $E \cap F = \{a \in A \mid a \in E \text{ and } a \in F\}$

COMPLEMENT $\overline{E} = \{a \in A \mid a \notin E\}$
(E^c)

A

$$p(\overline{E}) = 1 - p(E)$$



$$p(E \cup F) = p(E) + p(F) - p(E \cap F)$$

Ex Probability of rolling odd number or prime (with fair die)

$$E = \{1, 3, 5\}$$

$$p(E) = 3/6 = 1/2$$

$$F = \{2, 3, 5\}$$

$$p(F) = 3/6 = 1/2$$

$$E \cup F = \{1, 2, 3, 5\}$$

$$p(E \cup F) = 4/6 = 2/3$$

$$E \cap F = \{3, 5\}$$

$$p(E \cap F) = 2/6 = 1/3$$

$$p(E \cup F) = p(E) + p(F) - p(E \cap F)$$

$$2/3 = 1/2 + 1/2 - 1/3$$

Probabilities of poker hands

Pr III

Deck of 52 cards; 4 suits

spades hearts clubs diamonds
♠ ♥ ♣ ♦

13 ranks: 2-10, J, Q, K, A
jack queen king

How many different 5-card hands?

$$52 C_5 = \binom{52}{5} = \frac{52!}{5! 47!} = 2,598,960$$

HANDS

Straight flush: 5 cards of sequential rank,
all of same suit

Four of a kind: 4 cards of same rank

Full house: 3 cards of one rank,
2 cards of other rank

Flush: 5 cards of same suit
(not in sequence)

Straight: 5 cards of sequential rank
(not all same suit)

Three of a kind: 3 cards of one rank
2 cards of other ranks

Two pairs

2 cards of one rank
2 cards of other rank
(and one of 3rd rank)

A pair

2 cards of one rank,
3 cards of other ranks

FOUR OF A KIND

How many hands?

Choose RANK in 13 ways

Remaining card (kicker) can be chosen
in $(52 - 4) = 48$ ways

four-of-a-kind hands $13 \cdot 48 = 624$

Probability of getting four of a kind when dealt 5 cards

$$\frac{624}{\binom{52}{5}} \approx 0.00024 \quad (0.024\%)$$

TWO PAIRS

How many hands?

Choose 2 ranks for the pairs $\binom{13}{2}$

Once the rank is fixed, can choose
suits in $\binom{4}{2}$ ways - do this twice

Last card cannot be of these two ranks
so $52 - 2 \cdot 4 = 44$ choices left

two-pairs hands $\binom{13}{2} \binom{4}{2} \binom{4}{2} 44$

Probability of getting two pairs when dealt 5 cards

$$\frac{\binom{13}{2} \binom{4}{2} \binom{4}{2} 44}{\binom{52}{5}} = \frac{123,552}{2,598,960} \approx 0.0475 \quad (4.75\%)$$